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Forming device and forming method

Abstract

Provided is a forming device including: a lower die; an upper die for forming a laminated body along the surface shape of the lower die by pressing the laminated body against the lower die; a pushing force generating mechanism that generates pushing force for pressing the upper die against the lower die; and a control unit that controls the pushing force generating mechanism, the upper die has a plurality of forming members aligned in the longitudinal direction, the pushing force generating mechanism has a plurality of pushing force generating units connected to the plurality of forming members and configured to generate pushing force for pressing the forming members against the lower die, and the control unit controls the plurality of pushing force generating units so as to press a pair of forming members arranged adjacent in the longitudinal direction toward the lower die at different timings from each other.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of priority to Japanese Patent Application Number 2021-184792 filed on Nov. 12, 2021. The entire contents of the above-identified application are hereby incorporated by reference.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a forming device and a forming method for forming a laminated body of a plurality of laminated sheet materials containing reinforced fibers.

2. Description of Related Art

(3) Structural members of aircrafts or the like have various sectional shapes, and as a method for manufacturing such a structural member, there is a known method of pressing a laminated body of a plurality of laminated sheet materials containing reinforced fibers against a forming die and forming the laminated body to obtain an intended shape (for example, see Japanese Patent No. 5278790).

(4) Japanese Patent No. 5278790 discloses arranging a laminated body in a forming die and applying a pressure thereto by a bladder to form the laminated body along the forming die.

(5) When a laminated body is formed along a forming die, a wrinkle may occur in a particular portion such as a region having a large change in the shape of the forming die. Such a wrinkle is likely to occur particularly when the forming die has a curved surface shape having a curvature in the longitudinal direction and also has a curved portion including a concave shape or a convex shape in the width direction orthogonal to the longitudinal direction.

(6) When a pressure is applied by a bladder as disclosed in Japanese Patent No. 5278790, however, a region that is preferentially formed or a timing that the forming takes place is already determined in accordance with the positional relationship between the bladder and the forming die. It is therefore difficult to suitably adjust a region to preferentially perform forming, a timing to perform forming, or the like, such as by pressing a region where a wrinkle is likely to occur against a forming die prior to pressing other regions.

BRIEF SUMMARY

(7) The present disclosure has been made in view of such circumstances and intends to provide a forming device and a forming method that can suppress a problem of a wrinkle occurring in a laminated body when forming the laminated body along a forming die.

(8) A forming device for forming a laminated body of a plurality of laminated sheet materials containing reinforced fibers according to one aspect of the present disclosure includes a first forming die having a curved surface that extends in a longitudinal direction and includes at least any one of a concave shape and a convex shape with respect to a width direction; a second forming die configured to form the laminated body along a surface shape of the first forming die by pressing the laminated body against the first forming die; a pushing force generating mechanism configured to generate pushing force for pressing the second forming die against the first forming die; and a control unit configured to control the pushing force generating mechanism, wherein the second forming die has a plurality of forming members aligned in the longitudinal direction, wherein the pushing force generating mechanism has a plurality of pushing force generating units connected to the plurality of forming members and configured to generate pushing force for pressing the forming members against the first forming die, and wherein the control unit controls the plurality of pushing force generating units so as to press a pair of the forming members arranged adjacent in the longitudinal direction against the first forming die at different timings from each other.

(9) A forming method for forming a laminated body of a plurality of laminated sheet materials containing reinforced fibers according to one aspect of the present disclosure includes a fixing step of fixing one end of the laminated body to a first forming die having a curved surface that extends in a longitudinal direction and includes at least any one of a concave shape and a convex shape with respect to a width direction; and a forming step of forming the laminated body along a surface shape of the first forming die by pressing a second forming die having a plurality of forming members aligned in the longitudinal direction against the first forming die, wherein the forming step presses a pair of the forming members arranged adjacent in the longitudinal direction against

the first forming die at different timings from each other.

(10) According to the present disclosure, it is possible to provide a forming device and a forming method that can suppress a problem of a wrinkle occurring in a laminated body when forming the laminated body along a forming die.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) FIG. 1 is a perspective view illustrating a lower die and a laminated body according to a first embodiment of the present disclosure and illustrates a state before the laminated body is formed.

(2) FIG. 2 is a perspective view illustrating the lower die and the laminated body according to the first embodiment of the present disclosure and illustrates a state after the laminated body has been formed.

(3) FIG. 3 is a sectional view illustrating a forming device according to the first embodiment of the present disclosure and illustrates a state before an upper die starts motion toward the lower die.

(4) FIG. 4 is a right side view of the forming device illustrated in FIG. 3 and illustrates a state before the upper die starts motion toward the lower die.

(5) FIG. 5 is a block diagram illustrating a control configuration of the forming device according to the first embodiment of the present disclosure.

(6) FIG. 6 is a flowchart illustrating a composite material molding method using the forming device.

(7) FIG. 7 is a flowchart of a second forming step illustrated in FIG. 6.

(8) FIG. 8 is a sectional view illustrating the forming device in a lamination step of FIG. 6.

(9) FIG. 9 is a partial enlarged view of portion B of FIG. 8.

(10) FIG. 10 is a sectional view illustrating the forming device in the second forming step of FIG. 6 and illustrates a state where the upper die is moving toward the lower die.

(11) FIG. 11 is a right side view of the forming device illustrated in FIG. 10 and illustrates a state where the upper die is moving toward the lower die.

(12) FIG. 12 is a sectional view illustrating the forming device in the second forming step of FIG. 6 and illustrates a state where the upper die has completed the motion toward the lower die and stops.

(13) FIG. 13 is a right side view of the forming device illustrated in FIG. 11 and illustrates a state where the upper die has completed the motion toward the lower die.

(14) FIG. 14 is a sectional view illustrating a forming device according to a second embodiment of the present disclosure and illustrates a state before the upper die starts motion toward the lower die.

(15) FIG. 15 is a sectional view illustrating the forming device according to the second embodiment of the present disclosure and illustrates a state where the upper die is moving toward the lower die.

(16) FIG. 16 is a sectional view illustrating the forming device according to the second embodiment of the present disclosure and illustrates a state where the upper die has completed the motion toward the lower die.

DETAILED DESCRIPTION

First Embodiment

(17) A forming device **100** according to the first embodiment of the present disclosure and a forming method using the same will be described below with reference to the drawings. FIG. 1 is a perspective view illustrating a lower die **10** and a laminated body **200** according to the present embodiment and illustrates a state before the laminated body **200** is formed. FIG. 2 is a perspective view illustrating the lower die **10** and the laminated body **200** according to the present embodiment and illustrates a state after the laminated body **200** has been formed.

(18) The forming device **100** of the present embodiment is a device that forms the laminated body

200, which is a plurality of laminated sheet materials, along the surface shapes of the lower die (first forming die) **10** and an upper die (second forming die) **20**. As illustrated in FIG. 1, the laminated body **200** before formed is a planarly laminated multilayered sheet-like composite materials.

(19) In the present embodiment, as the laminated body **200**, a planarly shaped laminated body of a plurality of laminated reinforced fiber sheets (dry sheets) containing no matrix resin is used. When reinforced fiber sheets containing no matrix resin are used, a resin transfer molding (RTM) method is used in which the laminated body **200** formed along the surface shapes of the lower die **10** and the upper die **20** is arranged in a molding die (not illustrated) and a resin material is injected inside the molding die to impregnate the reinforced fibers therewith and mold the same. The reinforced fiber contained in the reinforced fiber sheet may be, for example, carbon fiber, glass fiber, aramid fiber, or the like.

(20) Although, as described above, a plurality of sheet-like reinforced fiber sheets containing no matrix resin are used as the laminated body **200** in the present embodiment, another manner may be possible. For example, a plurality of laminated reinforced fiber sheets containing a matrix resin may be used as the laminated body **200**.

(21) As the matrix resin contained in the reinforced fiber sheets, either a thermosetting resin material or a thermoplastic resin material can be used. The thermosetting matrix resin is, for example, an epoxy resin, an unsaturated polyester, a vinyl ester, a phenol, a cyanate ester, a polyimide, or the like.

(22) Examples of the thermoplastic matrix resin include polyetheretherketone (PEEK), polyethylene terephthalate (PET), polybutylene terephthalate (PBT), nylon 6 (PA6), nylon 66 (PA66), polyphenylene sulfide (PPS), polyetherimide (PEI), polyetherketoneketone (PEKK), and the like.

(23) When a thermoplastic resin is used as the matrix resin, the forming device **100** includes a heating mechanism (not illustrated) that can heat the thermoplastic resin contained in the laminated body **200** at the softening temperature or higher. By heating the thermoplastic resin at the softening temperature or higher by the heating mechanism, it is possible to form the laminated body **200** containing the thermoplastic resin along the surface shapes of the lower die **10** and the upper die **20**.

(24) Details of the forming device **100** according to the present embodiment will be described with reference to the drawings. FIG. 3 is a sectional view illustrating the forming device **100** according to the present embodiment and illustrates a state before the upper die **20** starts motion toward the lower die **10**. FIG. 4 is a right side view of the forming device illustrated in FIG. 3 and illustrates a state before the upper die starts motion toward the lower die. FIG. 5 is a block diagram illustrating a control configuration of the forming device **100** of the present embodiment. As illustrated in FIG. 3 to FIG. 5, the forming device **100** of the present embodiment includes the lower die **10**, the upper die **20**, an upper die (third forming die) **30**, a pushing force generating mechanism **40**, and a control unit **50**.

(25) The forming device **100** illustrated in FIG. 3 and FIG. 4 is arranged in a three-dimensional space. The X-axis, the Y-axis, and the Z-axis illustrated in FIG. 3 and FIG. 4 are axes intersecting each other in the three-dimensional space. The X-axis is an axis extending parallel to an installation surface S on which the lower die **10** is installed, and the Z-axis is an axis extending in a direction orthogonal to the installation surface S on which the lower die **10** is installed. The Y-axis is an axis orthogonal to both the X-axis and the Z-axis and extending in a sheet depth direction in FIG. 3.

(26) The lower die **10** is a block-shape die having a surface shape for forming the laminated body **200** and is made of a metal material, for example. The lower die **10** has an upper surface (first forming surface) **11**, a side surface (second forming surface) **12**, a convex surface (curved surface) **13**, a concave surface (curved surface) **14**, and a bottom surface **15** as the surface shape for forming the laminated body **200**. FIG. 3 is a sectional view taken along the arrow A-A of FIG. 4 and

illustrates a cross section of the lower die **10** at or near the middle part in the longitudinal direction LD parallel to the Y-axis.

(27) As illustrated in FIG. 3, the upper surface **11** of the lower die **10** is a surface extending planarly parallel to the X-axis. The side surface **12** of the lower die **10** is a surface extending planarly parallel to the Z-axis. The bottom surface **15** of the lower die **10** is a surface extending planarly in a direction intersecting the X-axis.

(28) The convex surface **13** is a surface connected between the upper surface **11** and the side surface **12** and has a circular arc shape whose normal direction of the surface gradually changes so as to change from a surface parallel to the X-axis to a surface parallel to the Z-axis as a position on the convex surface **13** approaches the side surface **12** from the upper surface **11** parallel to the X-axis. The convex surface **13** is a portion including a convex shape with respect to the width direction WD parallel to the X-axis.

(29) The concave surface **14** is a surface connected between the side surface **12** and the bottom surface **15** and has a circular arc shape whose normal direction of the surface gradually changes so as to change from a surface parallel to the Z-axis to a surface parallel to the X-axis as a position on the concave surface **14** approaches the bottom surface **15** from the side surface **12** parallel to the Z-axis and further change to a surface intersecting the X-axis. The concave surface **14** is a portion including a concave shape with respect to the width direction WD.

(30) As illustrated in FIG. 3, the laminated body **200** has a first end region **200a** and a second end region **200b** in the width direction WD. The second end region **200b** of the laminated body **200** is fixed to the lower die **10**. The second end region **200b** of the laminated body **200** is fixed to a predetermined position of the upper surface **11** of the lower die **10**.

(31) The shape of the lower die **10** illustrated in FIG. 3 may be another shape. For example, the upper surface **11** may be a surface extending in a direction different from the X-axis or may be a non-planar surface. Further, the side surface **12** may be a surface extending in a direction different from the Z-axis. Further, the convex surface **13** may have any convex shape different from a circular arc shape. Further, the concave surface **14** may have any concave shape different from a circular arc shape. Further, the bottom surface **15** may be a surface extending in a direction parallel to the X-axis or a direction inclined downward from the X-axis. The lower die **10** can be any shape including at least any one of a concave shape and a convex shape with respect to the width direction WD.

(32) The upper die **20** is a block-like die for pressing the laminated body **200**, the second end region **200b** of which is fixed to the lower die **10**, against the lower die **10** and forming the laminated body **200** along the surface shape of the lower die **10** and is made of a metal material, for example. The upper die **20** presses the laminated body **200** fixed to the lower die **10** against the side surface **12**, the convex surface **13**, the concave surface **14**, and the bottom surface **15** of the lower die **10** and forms the laminated body **200** along the surface shapes of the lower die **10** and the upper die **20**.

(33) As illustrated in FIG. 4, the upper die **20** has a plurality of forming members **21**, **22**, **23**, **24**, **25**, **26**, **27**, **28**, and **29** aligned adjacent to each other in the longitudinal direction LD. As illustrated in FIG. 4, the distances in the Z-axis direction from the forming members **21** to **29** to the bottom surface **15** of the lower die **10** are the same distance DO. That is, the distances from the forming members **21** to **29** to the bottom surface **15** of the lower die **10** before starting motion toward the lower die **10** are the same.

(34) As illustrated in FIG. 3, the forming member **21** has an under surface **21a**, a side surface **21b**, and a convex surface **21c** as the surface shape for forming the laminated body **200**. The under surface **21a** of the forming member **21** is a surface extending planarly in a direction intersecting the X-axis. The side surface **21b** of the forming member **21** is a surface extending planarly parallel to the Z-axis. The convex surface **21c** is a surface connected between the under surface **21a** and the side surface **21b** and has a circular arc shape whose normal direction of the surface gradually

changes so as to change from a surface intersecting the X-axis to a surface parallel to the Z-axis as a position on the convex surface **21c** approaches the side surface **21b** from the under surface **21a** parallel to the X-axis. The convex surface **21c** has a shape corresponding to the concave surface **14** of the lower die **10**.

(35) The forming member **21** has been described above, and since the same applies to the forming members **22**, **23**, **24**, **25**, **26**, **27**, **28**, and **29**, the description thereof will be omitted below. Note that the shape of the lower die **10** may have different shapes at respective positions along the longitudinal direction LD. When the shape of the lower die has different shapes at respective positions along the longitudinal direction LD, the shapes of the forming members **22**, **23**, **24**, **25**, **26**, **27**, **28**, and **29** will differ, respectively.

(36) As illustrated in FIG. 3, the upper die **30** is a block-like die for pressing a region at and near the second end region **200b** of the laminated body **200** against the upper surface **11**, which is adjacent to one side in the width direction WD of the convex surface **13** of the lower die **10**, and is made of a metal material, for example. The upper die **30** presses the laminated body **200** fixed to the lower die **10** against the lower die **10** to form the laminated body **200** along the surface shapes of the lower die **10** and the upper die **30**.

(37) The pushing force generating mechanism **40** is a mechanism that generates a pushing force for pressing the upper die **30** against the lower die **10**. As illustrated in FIG. 4, the pushing force generating mechanism **40** has pushing force generating units **41**, **42**, **43**, **44**, **45**, **46**, **47**, **48**, and **49**. As illustrated in FIG. 4, the pushing force generating units **41** to **49** are connected to the forming members **21** to **29**, respectively, and generate pushing force for pressing the forming members **21** to **29** against the lower die **10**.

(38) As illustrated in FIG. 3, the pushing force generating unit **41** has a fixing frame **41a**, a slide frame **41b**, and a drive unit **41c**. The fixing frame **41a** is a member installed above the upper die **30** and extending in the horizontal direction parallel to the X-axis.

(39) The slide frame **41b** is a member extending in the perpendicular direction parallel to the Z-axis, which is a member attached movably along the Z-axis relative to the fixing frame **41a**. The slide frame **41b** is made of aluminum, for example. It is preferable that the slide frame **41b** be a partially hollowed light weight structure. The lower end of the slide frame **41b** is connected to the forming member **21**.

(40) The drive unit **41c** has a motor, for example, and generates drive force for moving the slide frame **41b** along the Z-axis relative to the fixing frame **41a**. The drive unit **41c** moves the slide frame **41b** downward along the Z-axis and thereby presses the side surface **21b** of the forming member **21** against the laminated body **200** to form the laminated body **200** held between the side surface **21b** and the side surface **12** of the lower die **10**.

(41) Further, the drive unit **41c** further moves the slide frame **41b** downward along the Z-axis and thereby presses the under surface **21a** of the forming member **21** against the laminated body **200** to form the laminated body **200** held between the under surface **21a** and the bottom surface **15** of the lower die. Similarly, the drive unit **41c** presses the convex surface **21c** of the forming member **21** against the laminated body **200** to form the laminated body **200** held between the convex surface **21c** and the concave surface **14** of the lower die.

(42) The pushing force generating unit **41** has been described above, and since the same applies to the pushing force generating units **42**, **43**, **44**, **45**, **46**, **47**, **48**, and **49**, the description thereof will be omitted below. As illustrated in FIG. 4, the pushing force generating unit **42** has a fixing frame **42a**, a slide frame **42b**, and a drive unit **42c**. The pushing force generating unit **43** has a fixing frame **43a**, a slide frame **43b**, and a drive unit **43c**. The pushing force generating unit **44** has a fixing frame **44a**, a slide frame **44b**, and a drive unit **44c**.

(43) The pushing force generating unit **45** has a fixing frame **45a**, a slide frame **45b**, and a drive unit **45c**. The pushing force generating unit **46** has a fixing frame **46a**, a slide frame **46b**, and a drive unit **46c**. The pushing force generating unit **47** has a fixing frame **47a**, a slide frame **47b**, and

a drive unit **47c**. The pushing force generating unit **48** has a fixing frame **48a**, a slide frame **48b**, and a drive unit **48c**. The pushing force generating unit **49** has a fixing frame **49a**, a slide frame **49b**, and a drive unit **49c**.

(44) The control unit **50** is a device that controls the pushing force generating mechanism **40**. As illustrated in FIG. 5, the control unit **50** transfers control signals to the pushing force generating units **41** to **49** via signal lines **51**, respectively. The control unit **50** controls a plurality of pushing force generating units **41** to **49** so that a pair of forming members arranged adjacent in the longitudinal direction LD are pressed against the lower die **10** at different timings from each other, as described later.

(45) Next, a composite material molding method for molding a composite material by using the forming device **100** of the present embodiment to form the laminated body **200** will be described with reference to FIG. 6 to FIG. 13. FIG. 6 is a flowchart illustrating the composite material molding method using the forming device **100**. FIG. 7 is a flowchart of a forming step illustrated in FIG. 6. FIG. 8 is a sectional view illustrating the forming device **100** in a lamination step of FIG. 6.

(46) In the lamination step of step **S101**, a plurality of fiber sheets are laminated, and thereby the laminated body **200** is made. As illustrated in FIG. 8, in a state where a sub-die **60** is assembled to the lower die **10**, a horizontal surface is established by the upper surface **11** of the lower die **10** and an upper surface **61** of the sub-die **60**, and a plurality of reinforced fiber sheets are laminated one by one on the horizontal surface.

(47) FIG. 9 is a partial enlarged view of portion B of FIG. 8. As illustrated in FIG. 9, reinforced fiber sheets **201**, **202**, **203**, **204**, and **205** are laminated in this order on the upper surface **11** of the lower die **10** and the upper surface **61** of the sub-die **60**, and thereby a planar laminated body **200** is made. Note that the number of layers of the reinforced fiber sheets of the laminated body **200** may be any number.

(48) In a fixing step of step **S102**, the second end region **200b** of the laminated body **200** is fixed on the upper surface **11** of the lower die **10**. For example, the laminated body **200** is fixed to the lower die **10** by attaching a fixing member (not illustrated) to the upper surface **11** and holding the second end region **200b** between the fixing member (not illustrated) and the upper surface **11**. After the laminated body **200** is fixed to the lower die **10**, the sub-die **60** is removed from the lower die **10**.

(49) In a first forming step of step **S103**, the upper die **30** is pressed against the upper surface **11** adjacent to one side in the width direction WD of the convex surface **13** of the lower die **10** to form the laminated body **200**. Specifically, the upper die **30** is installed above the second end region **200b** of the laminated body **200** fixed to the upper surface **11** of the lower die **10**, the pushing force generating mechanism **40** is installed above the upper die **30**, and the second end region **200b** of the laminated body **200** is formed along the shape of the upper surface **11** by the weights of the upper die **30** and the pushing force generating mechanism **40**.

(50) Although the laminated body **200** is formed along the shape of the lower die **10** by the weights of the upper die **30** and the pushing force generating mechanism **40** herein, another manner may be possible. For example, the upper die **30** can also be structured such that the laminated body **200** is fixed to the lower die **10** to have a predetermined thickness by using a fastening structure. In such a case, by adjusting the predetermined thickness, it is possible to suitably adjust the forming force of the upper die **30** pressing the laminated body **200** against the lower die **10**.

(51) In a second forming step of step **S104**, the upper die **20** is pressed against the laminated body **200** downward along the Z-axis by the pushing force generating mechanism **40** to form the laminated body **200** along the surface shape of the lower die **10**. Herein, the second forming step will be described in detail with reference to FIG. 7.

(52) As illustrated in FIG. 7, in step **S201**, the control unit **50** controls the drive unit **41c** of the pushing force generating unit **41** to start motion of the forming member **21** toward the lower die **10** from the state illustrated in FIG. 3.

(53) In step **S202**, the control unit **50** controls the drive units **42c**, **43c** of the pushing force

generating units **42, 43** to start motion of the forming members **22, 23** toward the lower die **10** at a timing that a predetermined time has elapsed from the start of the motion of the forming member **21** toward the lower die **10**.

(54) In step **S203**, the control unit **50** controls the drive units **44c, 45c** of the pushing force generating units **44, 45** to start motion of the forming members **24, 25** toward the lower die **10** at a timing that a predetermined time has elapsed from the start of the motion of the forming members **22, 23** toward the lower die **10**.

(55) In step **S204**, the control unit **50** controls the drive units **46c, 47c** of the pushing force generating units **46, 47** to start motion of the forming members **26, 27** toward the lower die **10** at a timing that a predetermined time has elapsed from the start of the motion of the forming members **24, 25** toward the lower die **10**.

(56) In step **S205**, the control unit **50** controls the drive units **48c, 49c** of the pushing force generating units **48, 49** to start motion of the forming members **28, 29** toward the lower die **10** at a timing that a predetermined time has elapsed from the start of the motion of the forming members **26, 27** toward the lower die **10**. With the above operations, each motion of the forming members **21** to **29** toward the lower die **10** is started, and the state illustrated in **FIG. 10** and **FIG. 11** is obtained.

(57) **FIG. 10** is a sectional view illustrating the forming device **100** in the second forming step of **FIG. 6** and illustrates a state where the upper die **20** is moving toward the lower die **10**. **FIG. 11** is a right side view of the forming device **100** illustrated in **FIG. 10** and illustrates a state where the upper die **20** is moving toward the lower die **10**. **FIG. 10** is a sectional view taken along the arrow C-C of **FIG. 11**.

(58) As illustrated in **FIG. 10**, when the forming member **21** moves along the side surface **12** of the lower die **10** from above to below along the Z-axis, the laminated body **200** is pressed against the side surface **12** of the lower die **10** by the side surface **21b** of the forming member **21** to form the laminated body **200** along the shape of the side surface **12**.

(59) As illustrated in **FIG. 11**, the distances in the Z-axis direction from the forming members **21** to **29** to the bottom surface **15** of the lower die **10** are not the same. The distance in the Z-axis direction from the forming member **21** to the bottom surface **15** is **D1**, each distance in the Z-axis direction from the forming members **22, 23** to the bottom surface **15** is **D2**, and each distance in the Z-axis direction from the forming members **24, 25** to the bottom surface **15** is **D3**. Further, each distance in the Z-axis direction from the forming members **26, 27** to the bottom surface **15** is **D4**, and each distance in the Z-axis direction from the forming members **28, 29** to the bottom surface **15** is **D5**. As illustrated in **FIG. 11**, the distance **D2** is longer than the distance **D1**, the distance **D3** is longer than the distance **D2**, the distance **D4** is longer than the distance **D3**, and the distance **D5** is longer than distance **D4**.

(60) As illustrated in **FIG. 11**, the distances in the Z-axis direction from the forming members **21** to **29** to the bottom surface **15** of the lower die **10** are not the same, because timings to move respective forming members toward the lower die **10** have been differentiated. The control unit **50** controls the pushing force generating units **41** to **49** so that the timings that the forming members **22** to **29** arranged on the end side with respect to the center position in the longitudinal direction **LD** of the laminated body **200** are moved toward the lower die **10** are delayed from the timing that the forming member **21** arranged at the center position (predetermined position) in the longitudinal direction **LD** of the lower die **10** is moved toward the lower die **10**.

(61) Further, the control unit **50** controls the pushing force generating units **41** to **49** so that a plurality of forming members **22** to **29** are pressed against the lower die **10** sequentially from the center position in the longitudinal direction **LD** of the lower die **10** to the ends of the longitudinal direction **LD** of the laminated body **200** with the timings being sequentially delayed from the timing for the forming member **21**. The control unit **50** controls the pushing force generating units **41** to **49** to first press the forming member **21** against the lower die **10**, subsequently press the forming members **22, 23** against the lower die **10**, subsequently press the forming members **24, 25**

against the lower die **10**, subsequently press the forming members **26, 27** against the lower die **10**, and finally press the forming members **28, 29** against the lower die **10**.

(62) In step **S206**, the control unit **50** controls the pushing force generating unit **41** to stop the motion of the forming member **21** in response to establishment of a state where the laminated body **200** is held between the forming member **21** and the bottom surface **15** of the lower die **10**.

(63) In step **S207**, the control unit **50** controls the pushing force generating units **42, 43** to stop the motion of the forming members **22, 23** in response to establishment of a state where the laminated body **200** is held between the forming members **22, 23** and the bottom surface **15** of the lower die **10**.

(64) In step **S208**, the control unit **50** controls the pushing force generating units **44, 45** to stop the motion of the forming members **24, 25** in response to establishment of a state where the laminated body **200** is held between the forming members **24, 25** and the bottom surface **15** of the lower die **10**.

(65) In step **S209**, the control unit **50** controls the pushing force generating units **46, 47** to stop the motion of the forming members **26, 27** in response to establishment of a state where the laminated body **200** is held between the forming members **26, 27** and the bottom surface of the lower die **10**.

(66) In step **S210**, the control unit **50** controls the pushing force generating units **48, 49** to stop the motion of the forming members **28, 29** in response to establishment of a state where the laminated body **200** is held between the forming members **28, 29** and the bottom surface **15** of the lower die **10**. In response to the end of step **S210**, the state illustrated in FIG. **12** and FIG. **13** is established.

(67) FIG. **12** is a sectional view illustrating the forming device **100** in the second forming step of FIG. **6** and illustrates a state where the motion of the upper die **20** toward the lower die **10** has completed. FIG. **13** is a right side view of the forming device **100** illustrated in FIG. **12** and illustrates a state where the motion of the upper die **20** toward the lower die **10** has completed. FIG. **12** is a sectional view taken along the arrow D-D of FIG. **13**. With the above operations, the process of the second forming step of FIG. **6** ends.

(68) Although, in the above second forming step, the forming member **21** arranged at the center in the longitudinal direction LD out of the plurality of forming members **21** to **29** is moved toward the lower die **10** at the earliest timing, another manner may be possible. For example, when a portion having a large shape change in the longitudinal direction LD (a portion where a wrinkle is likely to occur) is present at a predetermined position in the longitudinal direction LD, it is preferable to move a forming member closest to that portion at the earliest timing toward the lower die **10**. In such a case, the control unit **50** performs control to first move the forming member close to the portion having a large shape change in the longitudinal direction LD, toward the lower die **10** and then move the forming members adjacent thereto toward the lower die **10**.

(69) For some shape of the lower die **10**, it may be suitable to move forming members toward the lower die **10** simultaneously at a plurality of positions (for example, adjacent positions) in the longitudinal direction LD in order to suppress a wrinkle. In such a case, the control unit **50** controls the pushing force generating units **41** to **49** to move two or more forming members of the forming members **21** to **29** simultaneously toward the lower die **10**.

(70) Further, it is preferable to store in advance, in a storage unit (not illustrated), timings to start motion of the plurality of forming members **21** to **29** suitable to suppress occurrence of a wrinkle in the laminated body **200** when forming the laminated body **200** by using the lower die **10**, the upper die **20**, and the upper die **30**. In such a case, in the second forming step of FIG. **6**, the control unit **50** reads the timings to start motion of the plurality of forming members **21** to **29** stored in the storage unit and controls the timings to start motion of the plurality of forming members **21** to **29**.

(71) In a resin injection step of step **S105**, the laminated body **200** formed along the surface shapes of the lower die **10** and the upper die **20** is arranged in a molding die (not illustrated). A resin material is then injected inside the molding die to impregnate the resin material into the plurality of reinforced fiber sheets of the laminated body **200**.

(72) The resin injection step may be performed by vacuum assisted resin transfer molding (VaRTM) in which only the lower die **10** is used, the upper surface of the laminated body **200** formed by the lower die **10** is covered with a vacuum bag film (not illustrated) to reduce the internal pressure, and then a resin is injected therein.

(73) In a curing step of step **S106**, the resin material impregnated into the plurality of reinforced fiber sheets of the laminated body **200** is cured. When the resin material is thermosetting, the resin material is heated at a curing temperature or higher to cure the resin material. When the resin material is a thermoplastic resin, the resin material is cooled at a temperature lower than a softening temperature to cure the resin material. With step **S101** to step **S106** described above, a composite material molding method for forming the laminated body **200** to mold a composite material by using the forming device **100** is performed.

(74) Effects and advantages achieved by the forming device **100** of the present embodiment described above will be described.

(75) According to the forming device **100** of the present embodiment, the upper die **20** is pressed against the lower die **10** having the convex surface **13** with respect to the width direction **WD**, and thereby the laminated body **200** is formed along the surface shape of the lower die **10**. The pushing force generating mechanism **40** is controlled by the control unit **50** to apply, to the upper die **20**, pushing force for performing the pressing against the lower die **10**.

(76) The upper die **20** has the plurality of forming members **21** to **29** aligned in the longitudinal direction **LD**, and the plurality of pushing force generating units **41** to **49** of the pushing force generating mechanism **40** are connected thereto, respectively. The control unit **50** controls the plurality of pushing force generating units **41** to **49** so that a pair of forming members arranged adjacent in the longitudinal direction **LD** are pressed against the lower die **10** at different timings from each other.

(77) According to the forming device **100** of the present embodiment, a pair of forming members arranged adjacent in the longitudinal direction **LD** are pressed against the lower die **10** at different timings from each other. When one of the pair of forming members forms the laminated body **200** held between the one and the lower die **10**, a clearance is formed between the other of the pair of forming members and the lower die **10**, and therefore, deformation of the laminated body **200** is tolerated. Thus, compared to a case where press molding using molds with an upper die of one-piece structure and a lower die of one-piece structure is performed or a case where all the plurality of forming members are pressed against the lower die **10** at the same timing in the forming device **100**, it is possible to suppress occurrence of a wrinkle which would otherwise be caused by not tolerating deformation of the laminated body **200**.

(78) According to the forming device **100** of the present embodiment, the timing to move the forming members **22**, **23**, which are arranged on the end side of the laminated body **200** with respect to a predetermined position in the longitudinal direction **LD** (for example, the center position), toward the lower die **10** is delayed from the timing to move the forming member **21**, which is arranged at the predetermined position, toward the lower die **10**, and thereby deformation of the laminated body **200** occurring when the forming members **21** to **29** are pressed against the lower die **10** can be propagated from the predetermined position to the ends of the laminated body **200**. Accordingly, it is possible to suppress occurrence of a wrinkle which would otherwise be caused by not tolerating deformation of the laminated body **200**.

(79) According to the forming device **100** of the present embodiment, the plurality of forming members **21** to **29** are pressed against the lower die **10** sequentially from the predetermined position to the ends of the laminated body **200** with sequentially delayed timings. Thus, deformation of the laminated body **200** occurring when the forming members **21** to **29** are pressed against the lower die **10** is propagated continuously from the predetermined position to the ends of the laminated body **200** without being retained between the lower die **10** and the upper die **20**, and occurrence of a wrinkle can be suppressed.

(80) According to the forming device **100** of the present embodiment, by pressing the laminated body **200** against the upper surface **11** adjacent to one side in the width direction WD of the convex surface **13** of the lower die **10**, it is possible to form a region on one side in the width direction WD of the laminated body **200** along the shape of the upper surface **11**. Further, by pressing the upper die **20** against the side surface **12** adjacent to the other side in the width direction WD of the convex surface **13** of the lower die **10**, it is possible to form a region on the other side in the width direction WD of the laminated body **200** along the shape of the side surface **12**. Further, it is possible to form the region interposed between the one side and the other side in the width direction WD of the laminated body **200** along the shape of the convex surface **13**.

(81) According to the forming device **100** of the present embodiment, by moving the upper die **20** along the side surface **12** of the lower die **10**, it is possible to form the laminated body **200** along the side surface **12** from a region close to the convex surface **13** to the end in the width direction WD while tolerating deformation of the laminated body **200**. Since the laminated body **200** is formed while deformation thereof is tolerated, occurrence of a wrinkle can be suppressed.

Second Embodiment

(82) Next, a forming device **100A** according to the second embodiment of the present disclosure will be described with reference to the drawings. The present embodiment is a modified example of the first embodiment, and except as specifically described below, the present embodiment is the same as the first embodiment, and duplicated description will be omitted below.

(83) FIG. **14** is a sectional view illustrating the forming device **100A** according to the present embodiment and illustrates a state before the upper die **20** starts motion toward the lower die **10**. FIG. **15** is a sectional view illustrating the forming device **100A** according to the present embodiment and illustrates a state where the upper die **20** is moving toward the lower die **10**. FIG. **16** is a sectional view illustrating the forming device **100A** according to the present embodiment and illustrates a state where the upper die **20** has completed the motion toward the lower die **10**.

(84) In the forming device **100** of the first embodiment, the pushing force generating mechanism **40** is to move the slide frame **41b** relative to the fixing frame **41a** installed above the upper die **20**. In contrast, a pushing force generating mechanism **40A** of the forming device **100A** of the present embodiment is to move a slide frame **41Ab** relative to a support frame **41Ad** installed on the installation surface S.

(85) As illustrated in FIG. **14** to FIG. **16**, the pushing force generating unit **41A** of the pushing force generating mechanism **40A** of the forming device **100A** has the slide frame **41Ab**, the support frame **41Ad**, and connection frames **41Ae** and **41Af**. The lower end of the slide frame **41Ab** is connected to the forming member **21**. A plate **70** for applying pushing force used for forming the laminated body **200** to the upper die **30** is arranged above the upper die **30**.

(86) The support frame **41Ad** is fixed to the installation surface S. The slide frame **41Ab** is fixed to the connection frames **41Ae**, **41Af**. The connection frames **41Ae**, **41Af** are attached to the support frame **41Ad** movably along the Z-axis. The slide frame **41Ab** moves in a direction along the Z-axis integrally with the connection frames **41Ae**, **41Af** in response to drive force being applied by the drive unit **41Ac**.

(87) Since the support frame **41Ad** is fixed to the installation surface S, the forming device **100A** of the present embodiment is reliably supported by the support frame **41Ad** so that the slide frame **41Ab** is not separated from the side surface **12** of the lower die **10**. Thus, when the upper die **20** is moved toward the lower die **10**, pushing force for pressing the laminated body **200** against the lower die **10** can be reliably applied.

(88) The forming devices described in the above embodiments are understood as follows, for example.

(89) The forming device according to the present disclosure is a forming device (**100**) for forming a laminated body (**200**) of a plurality of laminated sheet materials containing reinforced fibers, the forming device includes: a first forming die (**10**) having a curved surface (**13**) that extends in a

longitudinal direction (LD) and includes at least any one of a concave shape and a convex shape with respect to a width direction (WD); a second forming die (20) configured to form the laminated body along a surface shape of the first forming die by pressing the laminated body against the first forming die; a pushing force generating mechanism (40) configured to generate pushing force for pressing the second forming die against the first forming die; and a control unit (50) configured to control the pushing force generating mechanism, the second forming die has a plurality of forming members (21 to 29) aligned in the longitudinal direction, the pushing force generating mechanism has a plurality of pushing force generating units (41 to 49) connected to the plurality of forming members and configured to generate pushing force for pressing the forming members against the first forming die, and the control unit controls the plurality of pushing force generating units so as to press a pair of the forming members arranged adjacent in the longitudinal direction against the first forming die at different timings from each other.

(90) According to the forming device of the present disclosure, the second forming die is pressed against the first forming die having the curved surface including at least any one of a concave shape and a convex shape with respect to the width direction, and thereby the laminated body is formed along the surface shape of the first forming die. The pushing force generating mechanism is controlled by the control unit to apply, to the second forming die, pushing force for performing the pressing against the first forming die.

(91) The second forming die has a plurality of forming members aligned in the longitudinal direction, and a plurality of pushing force generating units of the pushing force generating mechanism are connected thereto, respectively. The control unit controls the plurality of pushing force generating units so that a pair of forming members arranged adjacent in the longitudinal direction are pressed against the first forming die at different timings from each other.

(92) According to the forming device of the present disclosure, a pair of forming members arranged adjacent in the longitudinal direction are pressed against the first forming die at different timings from each other. When one of the pair of forming members forms the laminated body held between the one and the first forming die, a clearance is formed between the other of the pair of forming members and the first forming die, and therefore, deformation of the laminated body is tolerated. Thus, compared to a case where a pair of forming members are pressed against the first forming die at the same timing, it is possible to suppress occurrence of a wrinkle which would otherwise be caused by not tolerating deformation of the laminated body.

(93) In the forming device according to the present disclosure, the plurality of forming members may include at least a forming member arranged at a predetermined position in the longitudinal direction and a forming member arranged on an end side of the laminated body with respect to the predetermined position, and the control unit may be configured to control the plurality of pushing force generating units so that a timing to move the forming member arranged on the end side toward the first forming die is delayed from a timing to move the forming member arranged at the predetermined position toward the first forming die.

(94) According to the forming device of the present configuration, the timing to move the forming member arranged on the end side of the laminated body with respect to a predetermined position in the longitudinal direction toward the first forming die is delayed from the timing to move the forming member arranged at the predetermined position toward the first forming die, and thereby deformation of the laminated body occurring when the forming members are pressed against the first forming die can be propagated from the predetermined position to the end of the laminated body. Accordingly, it is possible to suppress occurrence of a wrinkle which would otherwise be caused by not tolerating deformation of the laminated body.

(95) In the forming device of the above configuration, the control unit may be in a manner that controls the plurality of pushing force generating units so that the plurality of forming members are pressed against the first forming die sequentially from the predetermined position to the end of the laminated body with sequentially delayed timings.

(96) According to the forming device of the present aspect, a plurality of forming members are pressed against the first forming die sequentially from the predetermined position to the end of the laminated body with sequentially delayed timings. Thus, deformation of the laminated body occurring when the forming members are pressed against the first forming die is propagated continuously from the predetermined position to the end of the laminated body without being retained between the first forming die and the second forming die, and occurrence of a wrinkle can be suppressed.

(97) The forming device according to the present disclosure may include a third forming die (30) configured to press the laminated body against a first forming surface adjacent to one side in the width direction of the curved surface of the first forming die, and the pushing force generating mechanism may be configured to generate pushing force for pressing the second forming die against a second forming surface adjacent to the other side in the width direction of the curved surface of the first forming die.

(98) According to the forming device of the present configuration, by pressing the laminated body against the first forming surface adjacent to one side in the width direction of the curved surface of the first forming die, it is possible to form a region on one side in the width direction of the laminated body along the shape of the first forming surface. Further, by pressing the second forming die against the second forming surface adjacent to the other side in the width direction of the curved surface of the first forming die, it is possible to form a region on the other side in the width direction of the laminated body along the shape of the second forming surface. Further, it is possible to form the region interposed between the one side and the other side in the width direction of the laminated body along the shape of the curved surface.

(99) In the forming device according to the above configuration, the pushing force generating mechanism may be in a manner that forms the laminated body along the second forming surface by moving the second forming die along the second forming surface of the first forming die.

(100) According to the forming device of the present aspect, by moving the second forming die along the second forming surface of the first forming die, it is possible to form the laminated body along the second forming surface from a region close to the curved surface to the end in the width direction while tolerating deformation of the laminated body. Since the laminated body is formed while deformation thereof is tolerated, occurrence of a wrinkle can be suppressed.

(101) The forming methods described in the above embodiments are understood as follows, for example.

(102) The forming method according to the present disclosure is a forming method for forming a laminated body of a plurality of laminated sheet materials containing reinforced fibers, the forming method includes: a fixing step of fixing one end of the laminated body to a first forming die having a curved surface that extends in a longitudinal direction and includes at least any one of a concave shape and a convex shape with respect to a width direction; and a forming step of forming the laminated body along a surface shape of the first forming die by pressing a second forming die having a plurality of forming members aligned in the longitudinal direction against the first forming die, and the forming step presses a pair of the forming members arranged adjacent in the longitudinal direction against the first forming die at different timings from each other.

(103) According to the forming method of the present disclosure, the second forming die is pressed against the first forming die having the curved surface including at least any one of a concave shape and a convex shape with respect to the width direction, and thereby the laminated body is formed along the surface shape of the first forming die. The second forming die has a plurality of forming members aligned in the longitudinal direction.

(104) According to the forming method of the present disclosure, the forming step presses a pair of forming members arranged adjacent in the longitudinal direction against the first forming die at different timings from each other. When one of the pair of forming members forms the laminated body held between the one and the first forming die, a clearance is formed between the other of the

pair of forming members and the first forming die, and therefore, deformation of the laminated body is tolerated. Thus, compared to a case where a pair of forming members are pressed against the first forming die at the same timing, it is possible to suppress occurrence of a wrinkle which would otherwise be caused by not tolerating deformation of the laminated body.

(105) In the forming method according to the present disclosure, the plurality of forming members may include at least a forming member arranged at a predetermined position in the longitudinal direction and a forming member arranged on an end side of the laminated body with respect to the predetermined position, and the forming step may be configured to delay a timing to move the forming member arranged on the end side toward the first forming die from a timing to move the forming member arranged at the predetermined position toward the first forming die.

(106) According to the forming method of the present configuration, the timing to move the forming member arranged on the end side of the laminated body with respect to a predetermined position in the longitudinal direction toward the first forming die is delayed from the timing to move the forming member arranged at the predetermined position toward the first forming die, and thereby deformation of the laminated body occurring when the forming members are pressed against the first forming die can be propagated from the predetermined position to the end of the laminated body. Accordingly, it is possible to suppress occurrence of a wrinkle which would otherwise be caused by not tolerating deformation of the laminated body.

(107) In the forming method of the above configuration, the forming step may be in a manner that moves the plurality of forming members so that the plurality of forming members are pressed against the first forming die sequentially from the predetermined position to the end of the laminated body with sequentially delayed timings.

(108) According to the forming method of the present aspect, a plurality of forming members are pressed against the first forming die sequentially from the predetermined position to the end of the laminated body with sequentially delayed timings. Thus, deformation of the laminated body occurring when the forming members are pressed against the first forming die is propagated continuously from the predetermined position to the end of the laminated body without being retained between the first forming die and the second forming die, and occurrence of a wrinkle can be suppressed.

(109) In the forming method according to the present disclosure, the forming step may be configured to form the laminated body by pressing a third forming die against a first forming surface adjacent to one end in the width direction of the curved surface of the first forming die and pressing the second forming die against a second forming surface adjacent to the other side in the width direction of the curved surface of the first forming die.

(110) According to the forming method of the present configuration, by pressing the laminated body against the first forming surface adjacent to one side in the width direction of the curved surface of the first forming die, it is possible to form a region on one side in the width direction of the laminated body along the shape of the first forming surface. Further, by pressing the second forming die against the second forming surface adjacent to the other side in the width direction of the curved surface of the first forming die, it is possible to form a region on the other side in the width direction of the laminated body along the shape of the first forming surface. Further, it is possible to form the region interposed between the one side and the other side in the width direction of the laminated body along the shape of the curved surface.

(111) In the forming method according to the above configuration, the forming step may be in a manner that forms the laminated body along the second forming surface by moving the second forming die along the second forming surface of the first forming die.

(112) According to the forming method of the present aspect, by moving the second forming die along the second forming surface of the first forming die, it is possible to form the laminated body along the second forming surface from a region close to the curved surface to the end in the width

direction while tolerating deformation of the laminated body. Since the laminated body is formed while deformation thereof is tolerated, occurrence of a wrinkle can be suppressed.

Claims

1. A forming device for forming a laminated body of a plurality of laminated sheet materials containing reinforced fibers, the forming device comprising: a first forming die having a curved surface that extends in a longitudinal direction and includes at least any one of a concave shape and a convex shape with respect to a width direction; a second forming die configured to form the laminated body along a surface shape of the first forming die by pressing the laminated body against the first forming die; a pushing force generating mechanism configured to generate pushing force for pressing the second forming die against the first forming die; and a control unit configured to control the pushing force generating mechanism, wherein the second forming die has a plurality of forming members aligned in the longitudinal direction, wherein the pushing force generating mechanism has a plurality of pushing force generating units connected to the plurality of forming members and configured to generate pushing force for pressing the forming members against the first forming die, wherein the plurality of forming members includes at least a forming member arranged at a predetermined position in the longitudinal direction and a forming member arranged on an end side of the laminated body with respect to the predetermined position, and wherein the control unit controls the plurality of pushing force generating units to propagate deformation of the laminated body occurring when the forming members are pressed against the first forming die from the predetermined position to the end side of the laminated body by controlling a timing to move the forming member arranged on the end side toward the first forming die to be delayed from a timing to move the forming member arranged at the predetermined position toward the first forming die.
 2. The forming device according to claim 1, wherein the control unit controls the plurality of pushing force generating units so that the plurality of forming members are pressed against the first forming die sequentially from the predetermined position to the end of the laminated body with sequentially delayed timings.
 3. The forming device according to claim 1 further comprising a third forming die configured to press the laminated body against a first forming surface adjacent to one side in the width direction of the curved surface of the first forming die, wherein the pushing force generating mechanism generates pushing force for pressing the second forming die against a second forming surface adjacent to the other side in the width direction of the curved surface of the first forming die.
 4. The forming device according to claim 3, wherein the pushing force generating mechanism forms the laminated body along the second forming surface by moving the second forming die along the second forming surface of the first forming die.
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